

## LISBOA GEOFISICO, Portugal

WMO: 085350

Lat: 38.7191N

Lon: 9.1497W

Elev: 255

StdP: 14.56

Time Zone: 0.00 (EUW)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 42.2       | 44.3 | 23.7                          | 18.1 | 48.8 | 28.4 | 22.5 | 50.2 | 23.0                  | 57.3 | 20.7 | 55.1 | 8.5                      | 40   | 0.422 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 17.8                   | 91.8            | 68.2 | 88.4 | 67.5 | 85.2 | 66.8 | 72.1                | 84.7 | 70.2 | 82.1 | 69.0 | 80.0 | 7.5                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.9                            | 107.2 | 74.2 | 67.0 | 100.1 | 72.5 | 65.6 | 95.4 | 71.9 | 35.8          | 84.7 | 34.2 | 82.3 | 33.2 | 80.2 | 85.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 19.9              | 17.7 | 15.9 | DB                         | 38.8 | 99.1               | 3.1 | 3.6                                                | 36.6 | 101.6      | 34.8 | 103.7      | 33.0 | 105.7      | 30.8 | 108.3 |
|                   |      |      | WB                         | 33.7 | 74.3               | 3.7 | 3.5                                                | 31.0 | 76.8       | 28.8 | 78.9       | 26.8 | 80.8       | 24.1 | 83.3  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 63.7   | 53.6  | 54.7  | 58.0  | 61.1  | 65.9  | 70.7  | 73.2  | 74.8  | 71.6  | 66.8  | 58.8  | 54.3  | (6)  |
|                                                                        | DBStd       | 8.75   | 3.88  | 3.68  | 4.32  | 4.77  | 5.58  | 5.31  | 4.64  | 4.57  | 4.14  | 4.59  | 4.28  | 4.05  | (7)  |
|                                                                        | HDD50       | 33     | 13    | 7     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 10    | (8)  |
|                                                                        | HDD65       | 1611   | 352   | 290   | 220   | 138   | 54    | 6     | 0     | 0     | 0     | 30    | 190   | 331   | (9)  |
|                                                                        | CDD50       | 5031   | 126   | 138   | 251   | 334   | 493   | 622   | 720   | 769   | 648   | 521   | 265   | 144   | (10) |
|                                                                        | CDD65       | 1135   | 0     | 0     | 4     | 22    | 82    | 178   | 255   | 304   | 199   | 87    | 4     | 0     | (11) |
|                                                                        | CDH74       | 7024   | 0     | 0     | 21    | 103   | 550   | 1154  | 1732  | 2159  | 1003  | 301   | 1     | 0     | (12) |
|                                                                        | CDH80       | 2305   | 0     | 0     | 1     | 8     | 155   | 401   | 614   | 793   | 279   | 54    | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 7.7    | 7.1   | 7.9   | 8.1   | 8.2   | 8.4   | 8.1   | 9.0   | 7.9   | 7.0   | 6.8   | 6.9   | 7.4   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 28.9   | 4.1   | 3.6   | 2.8   | 2.4   | 1.9   | 0.7   | 0.1   | 0.2   | 1.2   | 3.7   | 4.8   | 4.0   | (15) |
|                                                                        | PrecMax     | 46.8   | 13.9  | 11.7  | 10.5  | 6.4   | 6.1   | 3.9   | 1.7   | 2.3   | 5.1   | 10.3  | 14.7  | 15.1  | (16) |
|                                                                        | PrecMin     | 16.4   | 0.0   | 0.0   | 0.0   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.2   | (17) |
|                                                                        | PrecStd     | 8.1    | 3.0   | 2.6   | 2.3   | 1.6   | 1.6   | 0.9   | 0.3   | 0.4   | 1.3   | 2.9   | 3.4   | 3.6   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 65.2  | 68.0  | 76.8  | 81.2  | 89.3  | 94.1  | 97.1  | 96.9  | 91.8  | 85.7  | 72.7  | 65.6  | (19) |
|                                                                        |             | MCWB   | 56.2  | 56.4  | 59.6  | 62.6  | 65.1  | 70.1  | 70.5  | 68.7  | 66.4  | 64.5  | 61.3  | 59.7  | (20) |
|                                                                        | 2%          | DB     | 62.5  | 64.8  | 72.3  | 76.8  | 84.3  | 88.9  | 91.0  | 91.4  | 86.5  | 80.3  | 69.3  | 63.4  | (21) |
|                                                                        |             | MCWB   | 56.5  | 55.1  | 57.2  | 61.9  | 64.7  | 68.2  | 68.5  | 68.5  | 65.6  | 64.1  | 61.2  | 58.9  | (22) |
|                                                                        | 5%          | DB     | 60.9  | 62.7  | 68.4  | 73.5  | 80.3  | 84.7  | 86.4  | 87.7  | 82.9  | 77.1  | 67.0  | 61.9  | (23) |
|                                                                        |             | MCWB   | 56.2  | 54.9  | 56.5  | 60.6  | 63.6  | 66.9  | 67.7  | 67.8  | 65.6  | 63.7  | 60.3  | 58.1  | (24) |
|                                                                        | 10%         | DB     | 59.5  | 60.6  | 65.2  | 69.6  | 76.1  | 80.7  | 82.4  | 84.1  | 79.5  | 74.1  | 65.1  | 60.5  | (25) |
|                                                                        |             | MCWB   | 55.4  | 54.2  | 55.7  | 58.7  | 62.1  | 65.9  | 66.6  | 67.2  | 65.0  | 62.9  | 59.4  | 56.8  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 60.6  | 60.2  | 63.0  | 66.5  | 70.6  | 75.2  | 74.5  | 75.4  | 71.7  | 69.7  | 66.4  | 62.3  | (27) |
|                                                                        |             | MCDB   | 61.7  | 62.6  | 70.4  | 75.2  | 80.5  | 85.5  | 85.8  | 84.7  | 84.0  | 75.4  | 68.4  | 63.5  | (28) |
|                                                                        | 2%          | WB     | 59.1  | 58.9  | 60.2  | 63.6  | 67.3  | 70.5  | 71.4  | 71.8  | 69.4  | 68.0  | 64.3  | 60.7  | (29) |
|                                                                        |             | MCDB   | 60.8  | 60.8  | 66.4  | 73.5  | 78.0  | 82.7  | 85.9  | 84.4  | 79.5  | 72.8  | 66.4  | 62.3  | (30) |
|                                                                        | 5%          | WB     | 57.9  | 57.6  | 58.8  | 61.3  | 65.0  | 68.6  | 69.4  | 70.0  | 68.2  | 66.6  | 62.6  | 59.3  | (31) |
|                                                                        |             | MCDB   | 59.7  | 60.3  | 63.4  | 70.6  | 75.9  | 80.9  | 81.9  | 81.8  | 76.9  | 71.9  | 65.6  | 61.2  | (32) |
|                                                                        | 10%         | WB     | 56.4  | 56.1  | 57.5  | 59.6  | 63.2  | 67.0  | 68.0  | 68.7  | 66.9  | 65.3  | 60.7  | 57.6  | (33) |
|                                                                        |             | MCDB   | 58.7  | 59.4  | 62.5  | 67.4  | 73.1  | 78.1  | 79.6  | 79.9  | 74.7  | 70.5  | 63.8  | 59.9  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.5  | 12.1  | 13.1  | 14.2  | 15.3  | 16.3  | 17.4  | 17.8  | 15.5  | 12.9  | 10.7  | 9.7   | (35) |
|                                                                        |             | MCDBR  | 11.4  | 14.2  | 17.9  | 19.1  | 21.1  | 22.5  | 23.1  | 22.2  | 20.1  | 17.0  | 12.0  | 10.2  | (36) |
|                                                                        |             | MCWBR  | 7.1   | 7.9   | 9.0   | 10.1  | 10.3  | 10.9  | 10.1  | 10.3  | 9.0   | 7.9   | 7.0   | 7.1   | (37) |
|                                                                        | 5% WB       | MCDBR  | 9.1   | 11.4  | 13.9  | 16.9  | 18.0  | 19.8  | 19.9  | 19.1  | 16.2  | 12.6  | 9.8   | 9.1   | (38) |
|                                                                        |             | MCWBR  | 7.3   | 8.0   | 8.2   | 9.8   | 10.3  | 10.7  | 10.0  | 9.9   | 8.3   | 7.0   | 7.1   | 7.4   | (39) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.328 | 0.341 | 0.370 | 0.387 | 0.395 | 0.399 | 0.384 | 0.397 | 0.396 | 0.362 | 0.338 | 0.328 | (40) |
|                                                                        | taud        |        | 2.516 | 2.476 | 2.404 | 2.350 | 2.358 | 2.365 | 2.400 | 2.373 | 2.388 | 2.503 | 2.527 | 2.515 | (41) |
|                                                                        | Ebn at Noon |        | 269   | 280   | 282   | 283   | 281   | 279   | 282   | 276   | 269   | 269   | 263   | 260   | (42) |
|                                                                        | Edh at Noon |        | 25    | 29    | 34    | 38    | 39    | 39    | 37    | 37    | 35    | 28    | 24    | 23    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 714   | 1031  | 1400  | 1802  | 2162  | 2334  | 2376  | 2156  | 1688  | 1162  | 794   | 644   | (44) |
|                                                                        | RadStd      |        | 89    | 120   | 147   | 145   | 141   | 121   | 93    | 82    | 106   | 95    | 97    | 72    | (45) |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (15 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page